CLOSED POINTS OF A DECENT SPACE WITH BOUNDED FIBRE DEGREES ARE GEOMETRICALLY CLOSED OVER A FIELD

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Editorial note. No omission markers in-span; the weak-decency input is delegated to an uncarded chapter lemma and declared external.

Allowed standard prerequisites (editorial).

- Residue fields and fibres. lemma-finite-residue-field-extension-finite (carded L26731). k'/k finite.
- Weakly decent presentations. lemma-weakly-decent-presentation (same chapter, NOT carded, declared external). The presentation is weakly decent.

Source reference. Original definition 03I8: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1262–1262. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03K5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2865–2865. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 03II: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 459–459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03M4: focused-seeds/geometry/S9-decent-decent-space-closed-point/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Suppose X is a decent algebraic space over S. Let $x \in |X|$ be a closed point. Then x can be represented by a closed immersion $i : \mathrm{Spec}(k) \to X$ from the spectrum of a field.*

Proof. We know that x can be represented by a quasi-compact monomorphism $i : \mathrm{Spec}(k) \to X$ where k is a field (Definition 03I8). Let $U \to X$ be an étale morphism where U is an affine scheme. As x is closed and X decent, the fibre F of

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$|U| \to |X|$ over x consists of closed points (Lemma 03K5). As i is a monomorphism, so is $U_k = U \times_X \mathrm{Spec}(k) \to U$. In particular, the map $|U_k| \to F$ is injective. Since U_k is quasi-compact and étale over a field, we see that U_k is a finite disjoint union of spectra of fields (Remark 03II). Say $U_k = \mathrm{Spec}(k_1) \amalg \ldots \amalg \mathrm{Spec}(k_r)$. Since $\mathrm{Spec}(k_i) \to U$ is a monomorphism, we see that its image u_i has residue field $\kappa(u_i) = k_i$. Since $u_i \in F$ is a closed point we conclude the morphism $\mathrm{Spec}(k_i) \to U$ is a closed immersion. As the u_i are pairwise distinct, $U_k \to U$ is a closed immersion. Hence i is a closed immersion (Morphisms of Spaces, Lemma 03M4). This finishes the proof. $\square$